

Logarithmic and exponential forms of equations



1

a $\log_2 32 = 5$

b $\log_6 36 = x$

c $\log_9 9 = 1$

d $\log_2 1 = 0$

e $\log_{0.9} 1 = 0$

f $\log_{16} 64 = 1.5$

g $\log_9 27 = \frac{3}{2}$

h $\log_{10} 0.01 = -2$

i $\log_3 \frac{1}{81} = -4$

j $\log_x 125 = 1.5$

2

a $4^2 = 16$

b $5^1 = 5$

c $8^0 = 1$

d $2^{-3} = 0.125$

e $(5.8)^2 = 33.64$

f $\left(\frac{1}{3}\right)^{-2} = 9$

g $x^5 = 32$

h $8^6 = x$

i $3^{-1} = \frac{1}{3}$

j $9^{\frac{1}{2}} = 3$

3 True

4 True

Inverse relationship between exponentials and logarithms

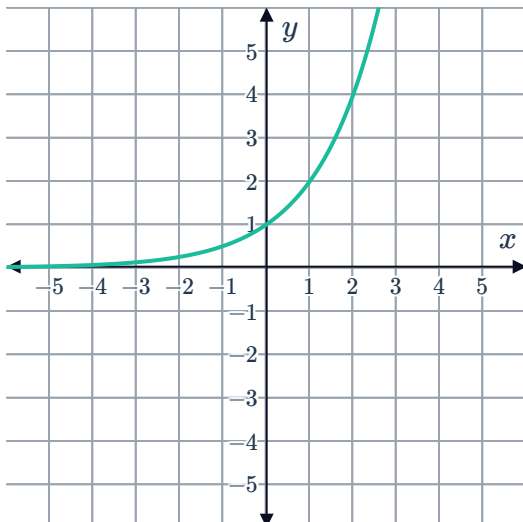
5

a Yes

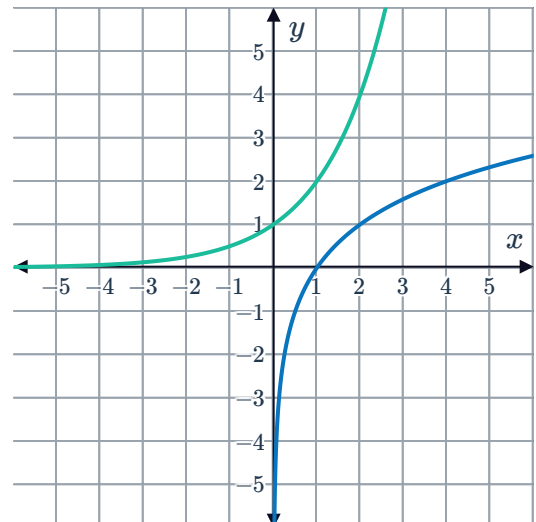
b $y = \log_{10} x$

6

a



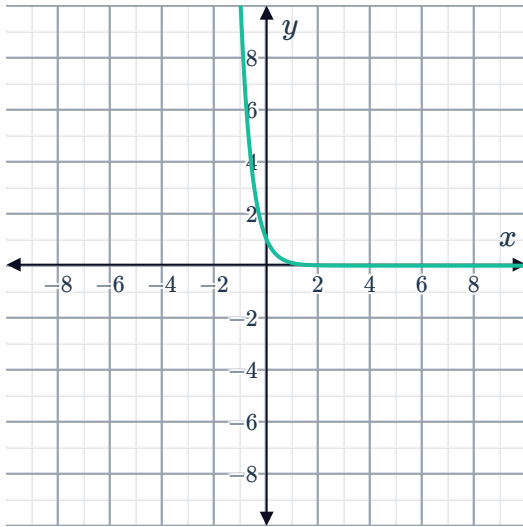
b





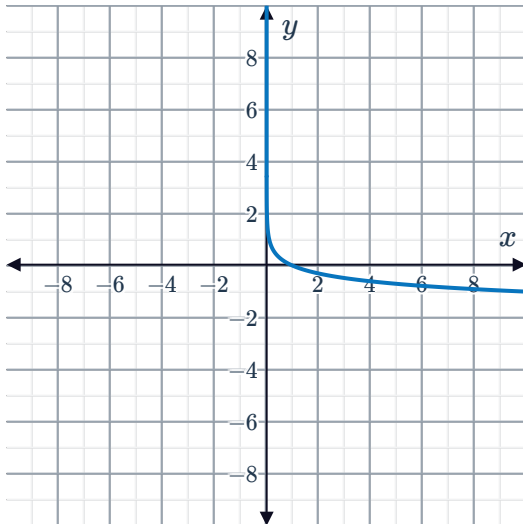
7

a



b $y = -\log x$

c



8

a $y = \log_e(x+3)$

b $x > -3$

c $(-\infty, \infty)$

9

a $y = 5^x + 3$

b $(-\infty, \infty)$

c $(3, \infty)$

10

a $y = \log_e\left(\frac{x+5}{2}\right)$

b $(-4.986, -4.986), (1.118, 1.118)$